

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 2 – Architecture Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Architecture Design Assignment
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	05-08-2026	Duration:	60 minutes

Code of Conduct

I B . Nandini (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. Nandini

Annexure A – Architecture Design Assignment

Agriculture Equipment Rental and Farm Resource Management Platform

Question

Based on your **assigned problem statement**, perform an **Architecture Design Analysis** and prepare an architecture design report by answering the following questions:

1. Analyze System Requirements

- Study the given problem statement and identify the system objectives.

1.1 System Objectives

The main objective of the Agriculture Equipment Rental and Farm Resource Management Platform is to provide an online platform where farmers can rent agricultural equipment from equipment owners and efficiently manage farm resources such as seeds, fertilizers, pesticides, labor, and expenses. The system also enables administrators to monitor users, equipment, bookings, and payments.

Objectives

- Enable online equipment rental.
 - Reduce equipment ownership costs.
 - Improve equipment utilization.
 - Manage farm resources digitally.
 - Support secure online payments.
 - Generate reports for administrators.
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- Analyze the functional and non-functional requirements that influence the software architecture.

1.2 Functional Requirements

- User Registration
- User Login
- Farmer Dashboard
- Equipment Owner Dashboard
- Admin Dashboard
- Equipment Listing
- Equipment Search
- Equipment Booking
- Booking Approval
- Booking Cancellation
- Online Payment
- Farm Resource Management
- Expense Management
- Notifications
- Reports

- Feedback & Ratings
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1.3 Non-Functional Requirements

Performance

- Response time should be less than 3 seconds.
- Support 1000+ concurrent users.

Security

- User authentication
- Encrypted passwords
- Secure online payment
- Role-based access control

Reliability

- Automatic backup
- Error recovery
- 99% uptime

Availability

- Available 24×7

Scalability

- Easy addition of users and equipment.

Maintainability

- Modular software architecture
- Easy software updates

- Identify key quality attributes such as scalability, maintainability, reliability, availability, performance, and security.

Quality Attributes

Quality Attribute	Description
Scalability	Supports increasing users and equipment
Maintainability	Easy to update modules
Reliability	Provides continuous service with minimal failures
Availability	System available anytime

Quality Attribute	Description
Performance	Fast search and booking
Security	Protects user and payment data

2. Select a Suitable Software Architecture

- Select an appropriate architectural style for the proposed system (e.g., Layered Architecture, Client-Server, MVC,

Layered Architecture (Three-Tier Architecture)

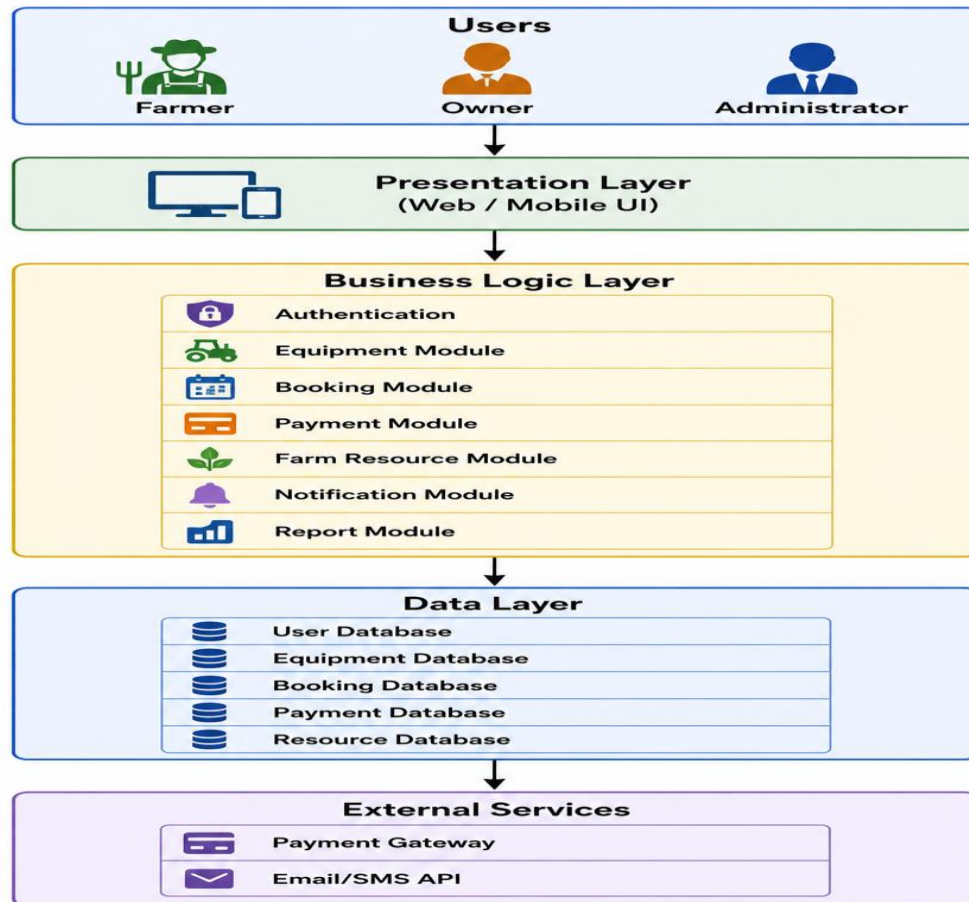
Layers

1. Presentation Layer
 2. Business Logic Layer
 3. Data Layer
- Microservices, Event-Driven, Service-Oriented Architecture, Serverless, or Hybrid Architecture).

Justification

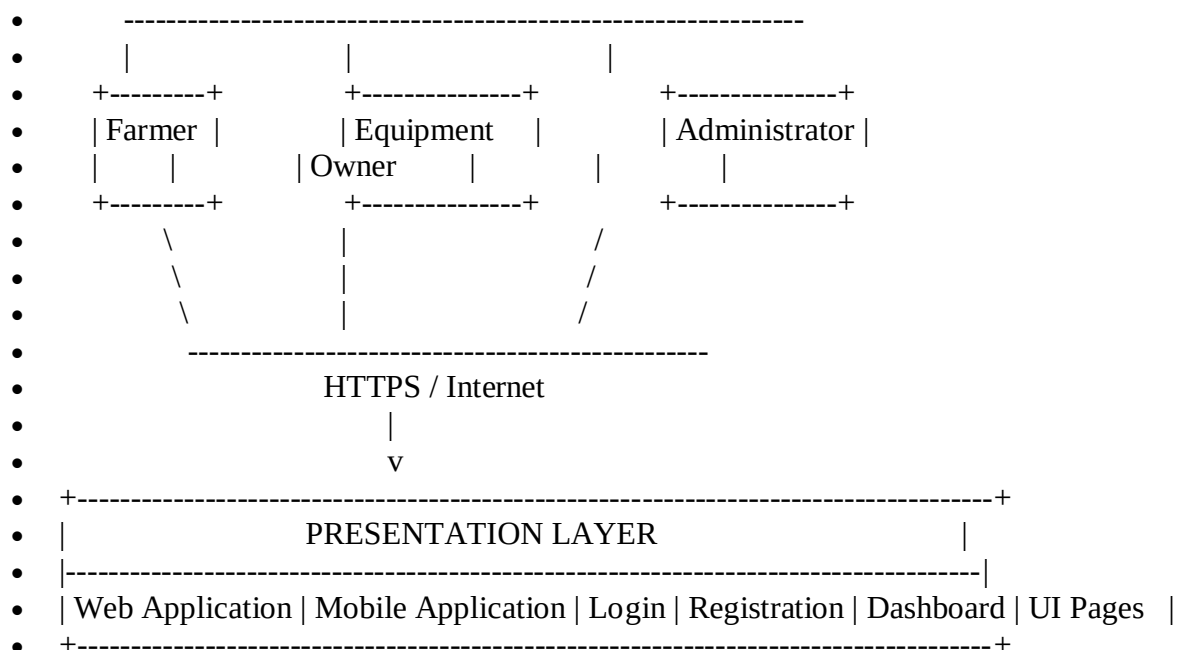
Layered Architecture is suitable because:

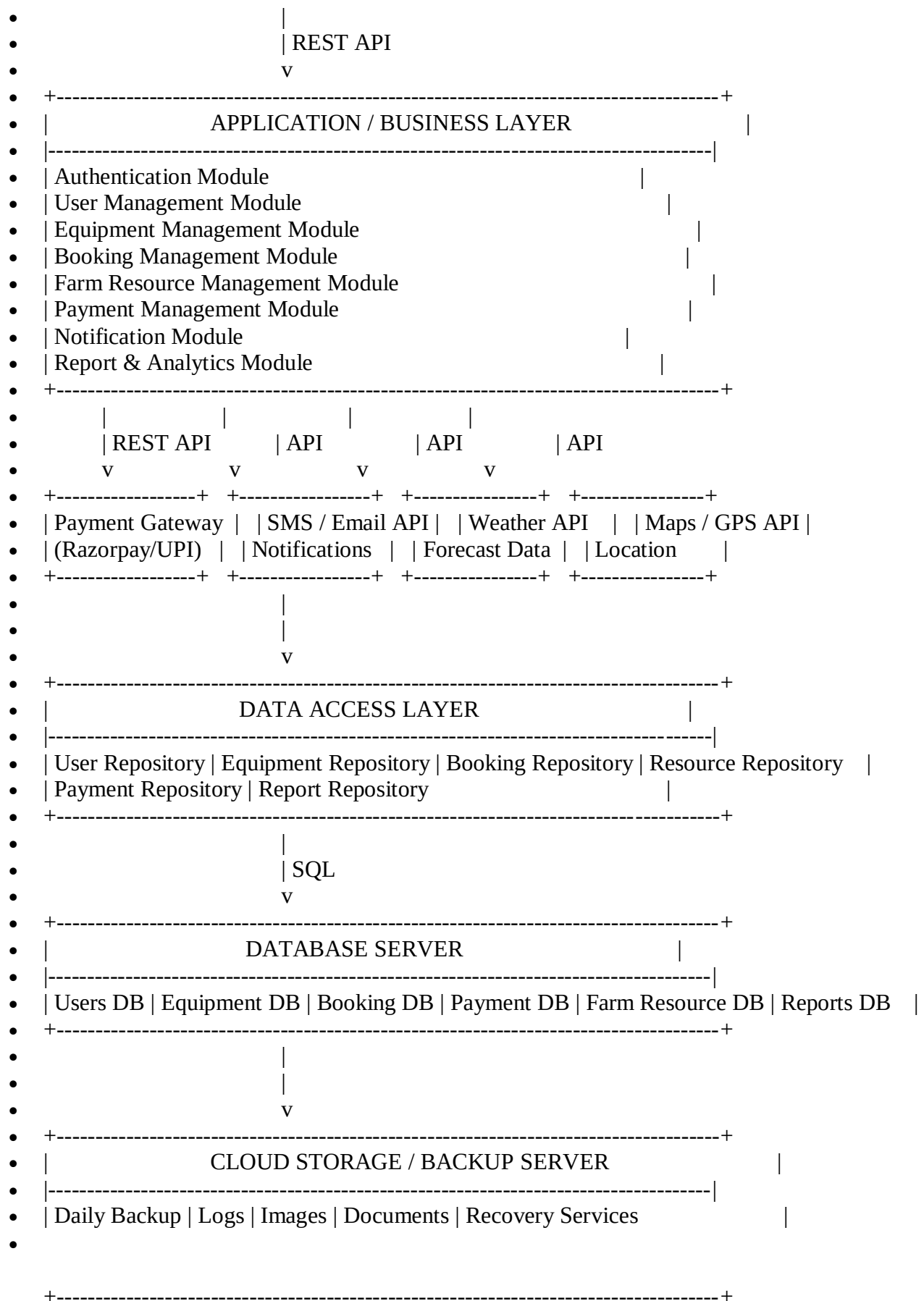
- Easy to develop and maintain.
 - Supports modular programming.
 - Provides better security.
 - Easy database management.
 - Easy future enhancement.
 - Commonly used for web applications.
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- Justify why the selected architecture is the most suitable for the given problem.



3. Draw the Software Architecture Diagram

- Design a clear architecture diagram illustrating the overall structure of the proposed system.
- USERS





- Include all major layers/components, databases, external systems, APIs, and communication mechanisms.

Source	Destination	Communication
Users → Presentation Layer	HTTPS	
Presentation Layer → Business Layer	REST API	
Business Layer → Payment Gateway	Secure Payment API	
Business Layer → SMS/Email Service	Notification API	
Business Layer → Weather API	REST API	
Business Layer → Maps/GPS API	REST API	
Business Layer → Data Access Layer	JDBC/ORM	
Data Access Layer → Database	SQL Queries	
Database → Cloud Backup	Secure Cloud Synchronization	

- Use appropriate software architecture notations and labels.

Major Components

- **Users:** Farmer, Equipment Owner, Administrator
- **Presentation Layer:** Web and Mobile User Interface
- **Business Layer:** Authentication, User Management, Equipment Management, Booking, Payment, Farm Resource, Notifications, Reports
- **External Systems/APIs:** Payment Gateway, SMS/Email API, Weather API, Maps/GPS API
- **Data Access Layer:** Repository classes for all modules
- **Database Server:** Users, Equipment, Booking, Payment, Farm Resources, Reports
- **Cloud Storage:** Backup and recovery services

4. Explain Components and Their Interactions

- Describe the purpose and responsibilities of each architectural component.

1. Presentation Layer

Responsibilities

- User Registration
- Login
- Equipment Search
- Equipment Booking
- Resource Management
- Payment Interface

2. Authentication Module

Responsibilities

- Verify user login
- Password encryption
- Session management

3. Equipment Module

Responsibilities

- Add Equipment
- Update Equipment
- Delete Equipment
- Display Equipment

4. Booking Module

Responsibilities

- Equipment booking
- Booking approval
- Booking cancellation

5. Payment Module

Responsibilities

- Online payment
- Payment verification
- Payment history

6. Farm Resource Module

Responsibilities

- Manage fertilizers
- Manage seeds
- Track expenses
- Track labor

7. Notification Module

Responsibilities

- Booking confirmation
- Payment confirmation
- Reminder notifications

8. Report Module

Responsibilities

- Booking reports
- Revenue reports
- User reports

9. Database

Stores

- Users
- Equipment
- Bookings
- Payments
- Farm Resources

- Explain how the components communicate and exchange data.

- **Users** (Farmer, Equipment Owner, Administrator) access the system through the **Web/Mobile Application** using **HTTPS**.
- The **Presentation Layer** sends user requests to the **Business Logic Layer** through **REST APIs**.
- The **Business Logic Layer** processes requests and communicates with the **Database** via the **Data Access Layer** using **SQL queries**.
- The Business Logic Layer also interacts with **Payment Gateway**, **SMS/Email API**, and **Weather API** to process payments, send notifications, and provide weather updates.
- The **Database** stores all user, equipment, booking, payment, and farm resource information, while **Cloud Storage** maintains secure backups.
- Illustrate the overall workflow of the system.

Workflow

1. Farmer logs into the system.
2. Searches for available equipment.
3. Books equipment.
4. Equipment owner approves the booking.

5. Farmer makes an online payment.
6. Booking details are stored in the database.
7. Notification is sent to both parties.
8. Administrator monitors all activities through reports.

5. Discuss Quality Attributes

Evaluate how the proposed architecture addresses the following aspects:

- Scalability
- New users can be added without affecting performance.
- Additional equipment categories can be introduced easily.

- Security

- Password encryption
- HTTPS communication
- Secure payment gateway
- Role-based access control

- Performance

- Fast equipment search
- Optimized database queries
- Caching frequently accessed data

- Reliability

- Automatic backups
- Error handling
- Database recovery

- Maintainability

- Modular architecture
 - Independent components
 - Easy bug fixing
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- Availability
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- 24×7 online access
 - Cloud deployment support
 - Backup server availability

6. Justify Architectural Decisions

- Explain the rationale behind your architectural choices.
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- Simple to understand and implement.
 - Separates user interface, business logic, and data storage.
 - Supports code reusability.
 - Improves testing and debugging.
 - Easy to maintain and extend.
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- Discuss the trade-offs considered while selecting the architecture.

Advantages

Easy maintenance
 Better security
 Modular design
 Easy testing

Disadvantages

Slightly slower due to multiple layers
 More communication between layers
 Additional development effort
 Can increase complexity for very small systems

- Explain how the chosen architecture supports future enhancements, system integration, and long-term maintainability.

Future Enhancements

- AI-based equipment recommendation.

- Weather forecasting integration.
- GPS tracking for equipment delivery.
- Mobile application (Android/iOS).
- Multi-language support.
- IoT integration for smart farming.
- Predictive maintenance for equipment.
- Chatbot support for farmers.

Conclusion

The **Layered (Three-Tier) Architecture** is the most suitable choice for the **Agriculture Equipment Rental and Farm Resource Management Platform** because it provides a clear separation of concerns, enhances security, supports scalability and maintainability, and allows future integration with external services such as payment gateways, weather APIs, GPS, and AI-based recommendation systems. This architecture satisfies both the functional and non-functional requirements of the project.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
System Requirement Analysis (30%)	Thoroughly analyzes the problem statement, accurately identifies functional and non-functional requirements, and clearly explains quality attributes influencing the architecture.	Analyzes most requirements with minor omissions or limited explanation.	Identifies only basic requirements with partial analysis.	Requirement analysis is incomplete, inaccurate, or lacks clarity.
Selection of Software Architecture (25%)	Selects the most appropriate architectural style with strong technical justification aligned to system requirements.	Selects a suitable architecture with reasonable justification.	Architecture is partially suitable with limited justification.	Inappropriate architecture selected or justification is missing.
Architecture Diagram and Component Design (20%)	Produces a clear, complete, and well-structured architecture diagram with all major components, interfaces, and interactions accurately represented.	Diagram is mostly complete with minor omissions or notation issues.	Diagram includes only essential components and lacks sufficient detail.	Diagram is incomplete, unclear, or technically incorrect.
Component Interaction and Quality Attribute Analysis (15%)	Clearly explains component responsibilities, interactions, and thoroughly discusses scalability, security, performance, reliability, maintainability, and availability.	Explains most components and quality attributes with minor gaps.	Provides limited explanation of interactions and addresses only some quality attributes.	Component interactions and quality attributes are poorly explained or missing.
Architectural Justification and Technical Presentation (10%)	Provides strong justification for architectural decisions, discusses trade-offs and future enhancements, and	Provides adequate justification with minor gaps; report is well organized.	Limited justification with minimal discussion of trade-offs or future scope;	Justification is weak or absent; report lacks organization and technical clarity.

	presents a well-organized, professional report.		report needs improvement.	
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Criteria	Mark
System Requirement Analysis (30%)	/5
Selection of Software Architecture (25%)	/5
Architecture Diagram and Component Design (20%)	/5
Component Interaction and Quality Attribute Analysis (15%)	/5
Architectural Justification and Technical Presentation (10%)	/5
TOTAL	/25